Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/15/2016
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	martin.volz@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	256-544-5078
Organization Name:	NASA Marshall Space Flight Center		
PI Address 1:	EM31		
PI Address 2:			
PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	10/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
Contact Email:	jan.r.rogers@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: end date change per P. Downey/MSFC (Ed., 1/12/2015) NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mazuruk, Konstantin (University of Alabama Huntsville) Cröll, Arne (University of Freiburg)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10%. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Progress:	Ground-based laboratory testing of ampoule configurations continued. The goal of the testing was to optimize the flight ampoule designs which must accommodate the variation in processing parameters in the flight experiment matrix. These parameters include the sample material (gallium-doped germanium or silicon-germanium) and the ampoule material (SiO2 or boron nitride). Determination of the internal ampoule components and their configuration is nearly complete. The science team has been working with the SCA (Sample Cartridge Assembly) thermal team to optimize the SCA configuration and furnace setpoints for the flight experiments. Thermal modelling has been done to help drive this optimization. Ground-based experiments with science ampoules in the ground LGF are expected to begin in 2016. An integrated design review (IDR) of the SCA and ampoule configurations is scheduled for May, 2016. An analysis of the dynamic stability of detached growth has been completed and is being prepared for publication. The stability model analyzes the response of a growing crystal to perturbations to its radius and crystal-melt interface. A linearized differential equation was used to analyze radius perturbations but a linear integro-diffential equation was required for the height perturbations. The stability model was applied to detached solidification under zero-gravity and terrestrial conditions. A numerical analysis was supplemented with an approximate analytical analysis, valid in the limit of small Bond numbers. For terrestrial conditions, a singularity was found to exist in the capillary stability coefficients where, at a critical value of the pressure differential across the meniscus, there is a transition from stability to instability. For the zero-gravity condition, exact formulas for the capillary stability coefficients were derived. Although the stability model was applied to the detached solidification crystal growth method, it can also be advantageously implemented to improve the stability analyses of other shaped crystal growth techniques.
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Abstracts for Journals and Proceedings	Volz MP, Mazuruk K. "Shape evolution of detached Bridgman crystals grown in microgravity." Presented at the 20th American Conference on Crystal Growth and Epitaxy, Big Sky, Montana, August 2–7, 2015. Abstracts. 20th American Conference on Crystal Growth and Epitaxy, Big Sky, Montana, August 2–7, 2015. , Aug-2015
Abstracts for Journals and Proceedings	Mazuruk K, Volz MP. "Determination of the contact angle based on the Casimir effect." Presented at the 20th American Conference on Crystal Growth and Epitaxy, Big Sky, Montana, August 2–7, 2015. Abstracts. 20th American Conference on Crystal Growth and Epitaxy, Big Sky, Montana, August 2–7, 2015. , Aug-2015

Abstracts for Journals and Proceedings	Volz MP, Mazuruk K, Cröll A. "Crystal Growth of Germanium-Silicon Alloys on the ISS." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, Virginia, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, Virginia, November 11-14, 2015. Abstracts. , Nov-2015